

STRATUS ET1 FUNCTIONAL SPECIFICATION

May 1, 1986

STRATUS ETL FUNCTIONAL SPECIFICATION

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NB: This document, written by Stratus Computer, Inc. staff, is based on an article by 'anon et al', published in the April, 1st, 1985 issue of Datamation, titled 'A Measure Of Transaction Processing Power'.

1. INTRODUCTION

In order to measure the performance of on-line transaction processing systems, a group of computer professionals (including academics, vendors, and users) created a benchmark profiling a banking transaction. The benchmark - also known as the debit-credit transaction, or ETL, (or mistakenly TPL, which it is not) has become an oft-quoted standard in the industry after it was used in 1973 by a large retail bank to select a vendor who could handle its 1,000 branches, 10,000 tellers, and 10,000,000 accounts at a peak load of 100 transactions per second.

In an effort to evaluate its systems, Stratus created the TPL benchmark. TPL - the Stratus-internal standard benchmark is not based on ETL, or the debit-credit transaction. The number and type of I/O operations it performs are different, and the database it operates on is different as compared to the one ETL operates on.

Due to various known and unknown reasons, it has become a popular misconception outside Stratus that TPL and ETL are the same. Vendor comparisons based on the TPL vs. ETL results, though done often, are invalid. In an attempt to clear up this misconception, and enable an 'apples to apples' comparison, Stratus is creating and benchmarking the ETL transaction as described in the article by 'anon et al' in the April, 1st, 1985 issue of Datamation.

The purpose of this document is to define the Stratus ETL benchmark, its profile, and the environment in which its performance is measured.

2. THE TRANSACTION

The definition of the Stratus ETL transaction is as follows:

Begin-Transaction

Read Input (X.25) Key = [Branch | Teller | Account]

Read Account (relative - keyed)
 Rewrite Account

 Read Teller (relative - random)
 Rewrite Teller

 Read Branch (relative - random)
 Rewrite Branch

 Write History (sequential)
 Write Acknowledgement (X.25)
 End-Transaction

One pair of Read and Rewrite I/O operations is equivalent to one Update operation.

3. THE DATABASE

The Stratus ETL benchmark will access four files. The attributes of these files are described below.

File Name	Record Size	File Size (per module)	Orgzn	Access
Account	100 bytes	200 MB	Relative	Indexed
Teller	100 bytes	0.2 MB	Relative	Random
Branch	100 bytes	0.02 MB	Relative	Random
History	50 bytes		Seqntl	Seqntl

File size refers to the size of files at the beginning of the benchmark.

These file sizes are one-fifth (1/5) the database size of the full implementation since the expected arrival rate of the Stratus ETL benchmark is twenty (20) transactions per second as compared to the hundred (100) transactions per second arrival rate of the full implementation (i.e. $20/100 = 1/5$). The final throughput of the system that meets response time requirements may be higher or lower than this arrival rate, twenty (20). Hence, this number twenty (20) should not be confused with the results of the benchmark.

There will be (per processing module) :

Two (2) million Account records
Two thousand (2000) Teller records
Two hundred (200) Branch records

The Account files will be a keyed access files (key length of ten (10) bytes), the other files will be random access files. All data files will have record level locking.

All keys will be generated randomly. The keys that are used as input to the benchmark will be generated such that eighty-five percent (85%) of the time the transaction is performed on a random account at the current branch, and fifteen percent (15%) of the time the transaction is performed on a random account at a different branch (which might be on a different module in the multi-module case).

4. RESULTS

In general, performance is quantified by measuring two metrics: response time, and transactions per second (tps). Response time is defined as the time taken to perform a single transaction (on a system under the full Stratus ETL benchmark load) after the arrival of the last bit of input from the communications line and before the transmission of the first bit of output to the communications line. Transactions per second (tps) is equal to the total transactions completed during the benchmark divided by the total elapsed time (in seconds) required to complete the benchmark. Hence, tps will include system overhead, line delays, X.25 time, etc.

The results will be from a system tuned to achieve the best performance. There are no restrictions on memory, or disks, or other tuning parameters.

The actual results of running this benchmark are reported separately. Since we expect these results to vary over time due to differences in operating system release, hardware configuration, etc., and we expect the ET-1 benchmark to remain the same, we have only functionally described the benchmark in this document.

C A R B O N C O P Y - - - - -

To: Clark Hodder, Franz Smolkovic
CC: Bill Elliott, Dick Luebke, Peter Kastner
Subject: Your comments on an Omri Serlin FTSN article
Date Sent: September 25, 1986 at 4:45 pm
From: Raphael Frommer

The debate on how we and Tandem ran ET-1 continues. Dick has just handed me the latest from Omri Serlin. Let me quote excerpts from the September 15th issue of his Fault Tolerant Systems Newsletter; it talks about the ET-1 methodology used by us and Tandem:

"the publicized Tandem results were obtained with a so-called 'lean' version of the benchmark, in which checkpointing between the primary and secondary TCPs was disabled. This fact was not disclosed."

Tandem notes that, despite the unprotected TCPs, its results are valid because auto-rollback was in effect during the timing."

My thoughts on the above statements, assuming they're both true:

1. TMF was clearly on; the database's integrity is guaranteed.
2. When Omri says checkpointing between primary and secondary TCPs was disabled, I am taking that to mean there were NO secondary TCPs running.
3. I believe Tandem used something in PATHWAY called TCP 'restart'. My impression is that under this mechanism the TCP sure enough dies completely, any transactions it had in progress are then aborted cleanly by TMF, and then (here's the cute part:) the PATHMON process, detecting the TCP's death, reconfigures all its terminals to a freshly started TCP. Of course, this results in the terminals coming up in their top level menu screen ... obviously not where they were when the original TCP first died. All the context that had been developed by the dead TCP is obviously lost. This "lean" mechanism is not as fault tolerant as possible because of this inevitable context loss.

I would like your thoughts on this subject, since you quit Tandem more recently than I and probably have a more current understanding of PATHWAY's capabilities. Am I indeed correct in saying that Omri's statements imply Tandem's ET-1 benchmark was not run as fault tolerant as it might have been?

Tandem has argued we cheated by not using transaction protection. Our counter has been that protection was not called for in the benchmark specification, and there is evidence to support that (the sequential logging file would serve such a purpose). We now may be able to counter claim that Tandem cheated. They boasted they ran a FT benchmark but didn't deliver FT requesters.

Stratus Computer, Inc. 07/09/86
 Configuration For XA600 Configuration for ET-1 Benchmark
 Module Name ml
 System Name

Model Number	Description	Unit Price (\$)	Qty	Total Price (\$)	Maint Price (\$)
-----SYSTEM-----					
P613	XA600 Package 8MB (8 Megabytes Duplexed) - Duplexed XA600 CPU's - Expansion Cabinet - Duplexed Memory Controller - Duplexed Communications Controller - 1 Communications Panel - 1 C103 Direct Async Line Adapter - Calendar Clock & Remote Support Link - Duplexed Power Supplies - VOS (Virtual Operating System)	263,900	1	263,900	806
T200	6250 TAPE CONTROLLER	10,000	1	10,000	85
T201	6250 TAPE DRIVE	25,000	1	25,000	200
C111	REMOTE SERVICE MODEM-S	0	1	0	0
-----DISK-----					
D119	448MB Disk Subsystem	39,000	2	78,000	408
D119-A	448MB Disk Add On	30,000	2	60,000	340
-----COMMUNICATIONS-----					
C109	SmartSync II	1,000	4	4,000	40
V102	Stratus Terminal	950	1	950	15
-----SOFTWARE-----					
S002	StrataNET	3,000	1	3,000	90
S006	TPF TRANSACTION PROC FACILITY	5,000	1	5,000	90
S008	FORMS MANAGEMENT SYSTEM (FMS)	1,500	1	1,500	40
S000	VOS	NA	NA	NA	120

TOTAL LIST PRICE : (\$) 451,350
 STRATUS INSTALLATION AND START-UP PACKAGE : (\$) 4,513
 TOTAL : (\$) 455,863
 MONTHLY MAINTENANCE CHARGE : (\$) 2,234

STRATUS XA600

To: Peter Kastner
CC: Clark Hodder, Larry Sherman, Bill Elliott
Subject: RE: ET1 "fault Tolerant newsletter"
Date Sent: November 4, 1986 at 9:38 pm
Answer From: Raphael Frommer

I know there is a debate on whether or not ET-1 REQUIRES protection.

One camp says:

- a. It's not specified in the benchmark (presumably as defined in the Datamation article).
- b. The sequential log file is written for purposes of recovery (with a flushing mechanism turned ON), and serves not only to recover from vendor-induced failures but can also support repair of databases corrupted by malformed applications. Vendor-supplied protection is consequentially unintended, uncalled for, and overkill.

Camp Two says:

The Datamation article does talk about transaction protection. Fig. 3 in it uses the phrases "Begin-Transaction" and "Commit-Transaction". The phrase "transactional file system" appears. Is "transactional" a superfluous word? No indeed; a vanilla file system becomes "transactional" when it can recognize transactions, which means locking and logging by transaction-id. Another sentence reads "All data files must be protected by fine granularity locking and logging."

It's possible for both camps to make hay out of the article, I'm afraid. For example, though the article does say "all data files must be protected by fine granularity locking and logging", the very next sentence admits that "the log file for transaction recovery must be duplexed to tolerate single failures". Camp Two is satisfied by the first sentence; Camp Two rejoices at the second.

My opinion:

At Stratus we've had a performance problem that forces us to turn off what I feel is implicit in the article, though not explicitly stated. Tprotection speed causes us to run in a mode that can't honestly be compared to Tandem's most frequently quoted "full function" performance figures. I understand that TMF is needed to shore up their FT inadequacies, but it's difficult to use that as an excuse. Consider that other non FT but scrupulous vendors would want to run this benchmark and would probably run with protection, if they have the product. Their technicians would do so as soon as they saw "Begin and Commit Transaction" in Figure 3. Turning it on wouldn't make them fault tolerant, merely compliant with the (poorly written) letter and spirit of the article.

Notwithstanding all of the above, I recognize we're between a rock and hard place. I for one continue to give customers the anti-tp argument based on the presence of the sequential log and hope they buy it. In my heart of hearts, I don't.

- - - C A R B O N C O P Y - - - - -

George Wilson
Raphael Frommer
Peter Kastner

(blind):
Subject: RE: ET1 "fault Tolerant newsletter"
Date Sent: November 5, 1986 at 9:39 am
Enclosure: tp_rebuttal
Answer From: Peter Kastner

George: There is some ambiguity on the ET-1 spec. Raphael's enclosed remarks state the case. I would duck the question as follows: Omri has objected to ALL vendor's incomplete or ambiguous ET-1 results reporting, and he's proposed a full disclosure mechanism in his newsletter. Due to fierce and misleading claims in the marketplace, Stratus is reluctant to be the first (and only) vendor to come 'clean' to Omri's spec. Therefore, while we would welcome a public discussion of Omri's proposed 'rules' for ET-1 reporting, we do not believe Omri's seminar is the proper place to debate whether and how much vendors 'cheated'.

We know, for example, that Tandem has extended the ET-1 response times from the clearly stated one second to 1.5 seconds at VLX announcement time to unlimited response time at the EXT announcement. Further, Tandem's frequently stated price/performance is based on configurations that do not support the disk storage necessary to hold the required ET-1 database (about 110MB per ET-1), thereby improving their price/performance.

Stratus has invited interested parties to watch us run ET-1 in Marlboro. This invitation stands.

In order to keep you out of the line of fire, George, you are authorized to present/explain any of the details in the June Stratus ET-1 white papers made available to the public. I don't think you should go beyond that.

Peter

directly to the FedEx local office.

Unfortunately, the scheme apparently underestimated the fact that virtually all large organizations are already equipped with high-speed fax machines, and have little incentive to use the service for internal document transfer. This left only small-to-medium size businesses lacking their own fax equipment, or those wishing to communicate with fax-less organizations, as candidates for ZapMail. The volume generated by these customers proved far smaller than projected, thus leading to the mounting losses.

This basic business miscalculation was aggravated by a slew of technical difficulties, including poor quality of the local dial-up phone lines connecting customer-installed machines to local FedEx offices; the company hoped to bypass the problem by giving customers direct access to its satellites (FTSN-44, p.19).

There was no public indication that the Tandem systems were in any way involved in the technical difficulties. The future of the Tandem systems in the ZapMail service remain uncertain at this point.

 *More DebitCredit information.* Chiming in with its own version of the DebitCredit benchmark, a small DEC VAR called ASCI (Advanced Systems Concepts Inc.) announced it obtained 14.9 tps @ 75% CPU utilization, and 9.8 tps @ 58% CPU utilization on a 28 MB DEC VAX 8650. Response time was at or below 0.7 seconds.

The benchmark was run using ASCI's own transaction management software, dubbed INTACT (Integrated Applications Control System). This software performs roughly the functions of DEC's TDMS and ACMS packages for terminal and application management, respectively. ASCI feels the DEC RA81 drive (a fixed drive with 458 MB, 36.3 ms average access, and 2.2 MB/s transfer rate; see FTSN-25, p.5) is the main limiting factor, peaking out at 35 I/Os per second. "We came away with a greater appreciation for the Tandem V8 disk drive", says ASCI.

The benchmarking effort, which was terminated after one week, was clearly no small undertaking. ASCI had been promising to supply the results to FTSN since August, 1985, but apparently was unable to do so until earlier this month.

Founded in July, 1981, Hoboken, NJ

based ASCI is headed by Ben Rosenberg. In addition to INTACT, it also offers a volume shadowing feature and a performance measurement tool for VAX systems, as well as consulting services. Its products are installed on a handful of VAX, PDP-11, and Professional 300 computers, especially in New York area banks, but also at Security Pacific bank.

Database Derby. You may be forgiven for never having heard of the Database Derby; but apparently 16 DBMS products took part in such an event, held in late August in San Francisco. We know about it because a Battelle Memorial Institute subsidiary called Information Dimension proudly announced its DM "advanced relational DBMS" had won the Business Rules Test, a part of the derby. Battelle has been trying to market the software package for several years now, with meager results: DM, which costs \$15K on supermicros and \$43.5K on mainframes, is installed at just over 60 sites.

CICS-on-PC battle heats up. Back in August, FTSN-48 p.11 reported on a PC-based CICS applications development facility being offered by Unicorn Systems. Now, a new competitor in this field has surfaced: Realia (Chicago, IL) offers its RealCICS applications development environment, which can run on either an IBM PC or a mainframe. Realia is headed by Marc Sokol.

Seminars and Conferences

27-30 October. 100th Presentation, Design Reliability Training Course. Syracuse, NY. \$850. Contact: Ms. Nan Pfrimmer, Reliability Analysis Center, Griffis AFB, NY 13441-5700, Telephone (315) 330-4151.

26-28 November. Multiprocessing and Parallel Systems. Conducted by O. Serlin, FTSN editor. Contact: Information Futures, Crays Pond House, Crays Pond, Reading, Berkshire RG8 7QG, U.K., 'phone 0491 681010, telex 848376.

8-11 December. Application of Reliability Tools for Designers. Orlando, FL. \$850. Contact: Ms. Nan Pfrimmer, Reliability Ana-

Transaction Previous

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----- CARBON COPY -----
!To: Bill Elliott
!CC: Dick Luebke, Jeff Kokes, John Kloos, Peter Kastner
!Subject: An embarrassing competitive situation. Advice needed.
!Date Sent: July 30, 1986 at 5:31 pm
!Options: Certified, Carbon Copy
!From: Raphael Frommer

Yesterday, at Hughes Aircraft Tucson I presented our published ET-1 performance figures to a crowd containing Stratus lovers as well as Tandem lovers. The most veteran, Tandem-knowledgeable member of the audience was armed with facts and figures, provided to him by Tandem. These included THEIR runs of ET-1, which he claimed were run by independent consultants hired by Tandem. He understood that our ET-1 benchmark profile did not include TPF protection, realized he could compare those only to Tandem's non-TPF results, but wanted TProtection-on data to compare to Tandem's TPF-on results. He claimed that the Tucson application would need to use protection in certain places and that some indication of our performance penalty was not an unreasonable request. I had no data to give him, obviously. I don't know what our allies thought of that answer, but, were I a Hughes employee in the audience, I could have drawn several different conclusions, none of which are favorable to Stratus:

1. Stratus views TProtection as so unimportant a feature that it simply didn't bother measuring its performance, and hence can't provide the data. Do I want to buy major amounts of gear from a company that doesn't measure what it sells, especially so important a transaction capability?

2. Stratus is no slouch. They measured TProtection performance, but the results are SO embarrassing, they refuse to divulge them.

Bill, I am expecting continued attacks along these lines from Tandem-philies. I can try to put our current TProtection performance in perspective:

1. It's a young product; you've got to expect lower performance than a product that's been maturing for years; we've got plans for improved performance in coming releases.

2. Our product is an especially thorough, paranoid product that takes no chance with your data; this implies heavy resource consumption.

3. TProtection is not needed to achieve transparent fault tolerance; TPF is vital on Tandem to that end. Do you really need such a mechanism, or are you being forced into using it to shore up a Tandem inadequacy?

4. Use control runoff as a higher performance alternative for certain applications not needing TProtection's full functionality.

Having made these points, and gained back some ground, I'll

mkktg#lib_02>Marketing>Pete_Kastner>waste_basket>mail.86-08-18.0001

still be asked, "So what is the performance hit?" It sounds ridiculous to say, "It's bad but we don't know exactly how bad, in ET-1 transactions per second."

Put yourself in the customer's position. What sort of an answer would you expect? What should I say in order not to lose technical creditability, either for myself or Stratus? I very much need some advice.

Stumped,
Raphael

COMPETITIVE POSITIONING

EXT10 and EXT25 systems compare very favorably with competitive systems in their respective classes. This section provides detail on competitive pricing and functionality.

First, we will focus on our pricing and performance advantages. Key measurements examined are entry-level price, transaction performance, and five-year cost of ownership. You will see that EXT systems are very competitive on a price and performance basis.

Next, we will focus on more qualitative measurements. We will discuss our competitors' selling strategies, their product capabilities, and how Tandem's advantages can help you win in competitive situations.

Price/performance comparison

EXT10 competitive lineup

	Tandem EXT10	IBM Sys 36	IBM Sys 38	IBM Sys 88	DEC 8200	Stratus 250
Hardware Entry Price	\$82.5 K	\$100 K	\$106 K	\$210 K	\$134 K	\$133 K
Performance (ET1 TPS)	4.3	1.4	2.0	.9	2.0	.9
5-Yr COO/TPS	\$36 K	\$117 K	\$99 K	\$345 K	\$118 K	\$254 K
Tandem's Performance Advantage		3.1 Times	2.2 Times	4.8 Times	2.2 Times	4.8 Times

- Notes:
1. The competition's configurations are equivalent to the base EXT10 system.
 2. Competitive TPS performance calculations were extrapolated from published performance information. (Stratus calculations do not include transaction monitoring.)
 3. The cost of ownership (COO) calculation includes 60 months of hardware maintenance and software fees (adjusted to reflect 15% cost of money).
 4. Tandem TPS performance information is based on the ET1 benchmark. See the "Product Features and Description" section for more details.
 5. Contact Phil Otis (Competitive Analysis) in Cupertino for detailed configuration information.